

HOPE ARTIFICIAL INTELLIGENCE

Classification Assignment

Chronic Kidney Disease (CKD) Prediction — Model Development & Evaluation

1. Problem Statement / Requirement

A requirement from the Hospital Management asked us to create a predictive model that will predict chronic kidney disease (CKD) based on several parameters. The client has provided the dataset for the same.

2. Problem Definition

The hospital wants a machine learning model that can predict whether a patient has chronic kidney disease (CKD) or not, based on medical test parameters such as blood pressure, sugar level, haemoglobin, creatinine, etc.

3. Dataset Overview

Total Rows: 399

Total Columns: 28

4. Data Pre-processing

The data was pre-processed by handling missing values and converting categorical/string values into numerical values using Label Encoding.

5. Model Development

The following machine learning algorithms were developed and evaluated to identify the best-performing classification model:

- Decision Tree Classifier
- Random Forest Classifier
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Naive Bayes

6. Evaluation Results

The performance of each algorithm was documented and compared using standard classification metrics:

Algorithm	Accuracy	Precision	Recall	F1-Score
Decision Tree	98%	98%	98%	98%
Random Forest	97%	97%	97%	98%
K-Nearest Neighbors (KNN)	88%	94%	96%	94%
Support Vector Machine (SVM)	98%	98%	99%	98%
Naive Bayes	96%	96%	99%	98%

Classification report for the final selected model (SVM):

	precision	recall	f1-score	support
0	0.96	1.00	0.98	45
1	1.00	0.97	0.99	75
accuracy			0.98	120
macro avg	0.98	0.99	0.98	120
weighted avg	0.98	0.98	0.98	120

7. Final Model Selection

Final Model: Support Vector Machine (SVM) — selected for achieving the highest and most balanced accuracy, precision, recall, and F1-score among all evaluated algorithms.

8. Submission Notes

Please note the following before final submission:

- Mail the completed document along with the .ipynb (code file) using the correct naming convention.
- Name the code file properly, e.g., **SVM classification assignment. ipynb**
- Ensure the document communicates the work clearly and professionally.
- Upload the project to GitHub and share the repository link.